
Learned Heuristics Decay With Distance to Goal: So Do Pattern Databases, by the Same Two-Moment Law

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Abstract

A heuristic is useful to a search only insofar as it orders the states that search compares. Wilt & Ruml (2016) measure this as Goal Distance Rank Correlation: one Kendall τ against exact distance-to-goal, pooled over every pair. We measure it *stratified by distance* instead, against exhaustive ground truth, on Rubik’s-cube sub-problems from 5.5×10^2 to 3.1×10^{23} states and on sliding-tile boards.

Call the states at exact distance d the *shell* at d . Search compares adjacent shells, and a heuristic’s accuracy there is $\Phi(\Delta/(\sigma\sqrt{2}))$, for Δ the gap in mean heuristic value between shells and σ the spread within them. Nothing is fitted. On disjoint halves of each shell it gives mean absolute error 0.012 over 252 observations against a noise floor of 0.002, and it survives a $300\times$ larger sub-problem, three further objectives, and injected noise.

Three consequences. Ordering quality is invariant to rescaling. Decay with distance occurs exactly when σ grows, so it is *not* particular to learned heuristics: pattern databases on the same states decay as much, in both domains, and matched on strength the two classes are indistinguishable. And σ is the one term an objective can act on, yet every objective built to act on it produced a worse heuristic.

The second domain bounds the law rather than confirming it. Error is nearly four times larger on the sliding tile, but that is not a fact about the state space: the error tracks within-shell skewness, monotonically across three tasks, and matched on skew the domains agree below $|\text{skew}| = 1$. The law is skew-limited, not domain-limited.

We also show τ -b rewards coarse-valued heuristics by enough (0.084) to reverse comparisons it is used for. The identity is classical; the measurement against exact ground truth is ours. We report three claims from earlier versions that our own controls refuted.

1 Introduction

Learned heuristics for combinatorial search are usually judged by what a search does with them: solve rate, solution length, nodes expanded. Each of those is a property of the triple (heuristic, search algorithm, budget), not of the heuristic. Change the beam width and every one of them moves. This makes them poor instruments for the question practitioners actually face, which is whether a trained heuristic is any good before committing compute to a search.

There is an established answer. Chrestien et al. (2023) prove that forward search with $f(s) = \alpha g(s) + \beta h(s)$ is strictly optimally efficient if and only if h is a perfect ranking, and conclude that “the absolute value of the heuristic is not important”, an observation credited to Xu et al. (2009) for beam search. Wilt & Ruml (2016) turn this into a measurement: Goal Distance Rank Correlation, Kendall’s τ between h and the exact edge-count distance d^* , computed offline from a backward breadth-first search. GDRC is search-independent, predicts greedy best-first expansions, and can be optimised directly to construct heuristics.

Our starting point is that GDRC is a *pooled* statistic, and that pooling discards a structure the heuristics it summarises actually have: ordering quality varies with distance from the goal. We expected that structure to be peculiar to bootstrapped heuristics. Deep Approximate Value Iteration (Agostinelli et al., 2019)

anchors its value estimate at the goal and propagates outward, so error should accumulate with distance by construction, and we report that this expectation is wrong. Abstraction-based heuristics decay too, and once heuristic strength is controlled the two classes are indistinguishable. What survives is the measurement problem rather than the mechanism story: a single coefficient averaged over all states cannot express a distance-dependent profile, and is dominated by state pairs that are far apart, which are easy to order and which no search ever compares.

Contributions.

1. A family of Rubik’s-cube sub-problems in which state-space cardinality can be varied over 21 orders of magnitude with branching factor, encoding, architecture and training budget held fixed, and in which *depth can be varied at exactly constant cardinality* by restricting the generating set (Section 3). Exhaustive ground truth is available on the small rungs; a meet-in-the-middle prover, verified against that ground truth, extends it by one further rung.
2. The measurement: ordering accuracy against exact ground truth, stratified by true distance, and a strength-controlled comparison showing that the decay it reveals belongs to informative heuristics in general, not to learned ones (Section 4). This is a negative result against our own initial hypothesis, and Section 4 states plainly what the earlier, under-controlled version of the comparison claimed.
3. A parameter-free account of that profile: per-shell ordering accuracy is $\Phi(\Delta/(\sigma\sqrt{2}))$ in the between-shell gap and within-shell spread, verified in and out of sample and across heuristic classes (Section 5). It explains why the decay is universal, shows that ranking quality is invariant to rescaling, and identifies within-shell spread as the single term an objective can act on. We claim the measurement, not the identity, which is standard signal detection theory.
4. A negative result on the intervention the law most obviously suggests (Section 6). Penalising within-shell spread is inert at low weight and monotonically harmful at high weight, because it stalls the depth curriculum; a ranking loss restricted to adjacent shells produces the flattest profile we measured and one of the worst heuristics. Every objective that flattened the profile degraded the search, which we report as a caution against reading the law as a design target.
5. An exact decomposition of pooled τ into that profile, which makes the dilution argument an identity rather than an intuition, and a measurement of a bias in τ -b that favours coarse-valued heuristics over continuous ones by about 0.084, enough to reverse comparisons of the kind this statistic is routinely used for (Section 7).
6. Evidence that the pooled statistic is diluted by pairs search never makes, including a run whose GDRC is indistinguishable from a working classical heuristic while its adjacent-shell ordering is near chance (Section 7).
7. A depth-versus-size experiment at constant cardinality, showing that size is not the operative variable in that design, and, at ten seeds per condition, that depth is not either over the range we can hold cardinality fixed (Section 8). A second correction lives here: the apparent failure at the largest diameter, which an earlier version of this work reported as a depth effect, is a curriculum-gate deadlock occurring in 1 run of 40 independently of diameter. We give the one-integer diagnostic that distinguishes the two.

2 Background and related work

The nearest prior work. Wilt & Ruml (2016) define GDRC and validate it as a predictor of search cost, reporting that “when the GDRC is below roughly 0.4, greedy best-first search performs very poorly.” Everything we might claim about *a ranking measure against exact ground truth* is theirs: it is search-independent, ranking-based rather than magnitude-based, predictive, and used prospectively to build heuristics. What remains is that GDRC is one scalar per domain; their Algorithm 1 samples nodes from a backward BFS and returns a single τ . Our contribution is that same quantity resolved by distance. We note that this is

not only a correction applied to learned heuristics: the pattern databases GDRC was designed around have distance-dependent profiles too (Section 4), so the pooling loss applies within its original scope.

Predicting search cost is not new. The Korf–Reid–Edelkamp formula (Korf et al., 2001) predicts IDA* expansions from the heuristic’s cumulative *value* distribution to within 1% on Rubik’s Cube. It is provably ranking-blind: two heuristics with identical value histograms but different state-to-value assignments receive identical predictions. It also constrains what we may claim. KRE prove the asymptotic heuristic branching factor equals the brute-force branching factor, so cost scales with problem growth; *we therefore make no claim that required search budget is independent of state-space size*, and our depth experiment is designed only to rule size out within a family where it is held constant.

Known failure modes, and how ours differs. Hadar et al. (2026) diagnose train/deploy distributional shift in DAVI: search oversamples low-heuristic states, causing systematic underestimation where search operates. That is induced by the search’s own sampling; our profile is measured on states drawn independently of any search. Hernández & Baier (2012) name “heuristic depression”, a *bounded region* trapping real-time search; what we measure is monotone in distance across the whole space. Czechowski et al. (2021) quantify how a learned value function’s ordering degrades with the *step spacing between compared states* and report a signal-to-noise ratio at one fixed spacing, the closest quantitative statement to ours, differing in the conditioning variable. Zawalski et al. (2024) observe qualitatively that value functions fail far from the goal, but probe this by injecting distance-independent Gaussian noise, without ground-truth comparison or stratification.

Scaling and capacity. Pendurkar et al. (2023) show the parameters needed to approximate a heuristic grow exponentially in instance size and prove a complexity-theoretic accuracy barrier. Their sweep varies instance size with branching factor changing alongside; ours holds branching, encoding, architecture and budget fixed. Futuhi & Sturtevant (2026) publish ordinal value losses, PAC-dimension bounds and *near*-admissible learned heuristics with nonzero measured overestimation. Arfaee et al. (2011) bootstrap from backward random walks of increasing length, which is the depth curriculum we adopt and claim nothing for.

3 Experimental setup

Notation. Three terms recur. A *shell* is the set of states at a given exact distance from the goal; shell d and shell $d + 1$ are one move apart, and those are the states a search compares against each other. A *rung* is one member of the family of sub-problems below, indexed by how many pieces it tracks; higher rungs are exponentially larger. And $d^*(s)$ denotes the exact number of moves from state s to the goal, obtained by exhaustive search rather than estimated. Φ is the standard normal distribution function throughout.

A ladder of sub-problems. We track k of the 24 edge “wing” pieces of a 4×4 Rubik’s cube and treat the remainder as interchangeable. This is exactly the projection a pattern database uses, so each rung is a genuine sub-problem: a move acts deterministically on the record of where the tracked pieces are. Rung k has $P(24, k) = 24!/(24 - k)!$ reachable states, from 552 at $k = 2$ to 3.10×10^{23} at $k = 24$. Every rung uses the same 63 generators, the same 24 slots and the same 600-wide one-hot encoding (24 slots $\times$ 25 symbols, the 25th being “not tracked”), so only cardinality varies. Reachable counts were verified against $P(24, k)$ by exhaustive search where feasible.

Exact ground truth. Breadth-first search enumerates rungs to $k = 6$ (96,909,120 states, 191 MB, ~ 22 minutes). Table 1 gives the resulting diameters and mean distances. Diameter is exactly $k + 2$ ($R^2 = 1.00000$) and mean distance is $0.864k + 1.242$ ($R^2 = 0.99988$), so across the ladder cardinality grows by 21 orders of magnitude while depth grows linearly. Depth is therefore measured, not assumed, and cannot alone explain a collapse of that shape.

Beyond enumeration. A meet-in-the-middle prober gives exact distances for a *sample* on larger rungs: build the ball of radius f around the goal once, then walk b levels out from each query state and intersect. Before use it is checked against exhaustive ground truth wherever both can run: 200/200 agreement at $k = 4$

k	states	diameter	mean distance
2	552	4	2.96
3	12,144	5	3.83
4	255,024	6	4.71
5	5,100,480	7	5.58
6	96,909,120	8	6.40

Table 1: Exhaustive ground truth. Diameter is exactly $k + 2$.

and 300/300 at $k = 6$. It reaches one rung beyond enumeration ($k = 8$); Section 10 explains why it stops there.

Varying depth at constant size. Restricting the generating set leaves the group generated unchanged (a half turn is two quarter turns), so the reachable set is identical while the diameter grows. On rung $k = 4$, four generating sets all reach exactly 255,024 states (verified by exhaustive search) at diameters 6, 8, 10 and 12. The cost of this design is that fewer generators also means a smaller branching factor: it varies depth and branching together and can rule out size, nothing more.

Training. Deep Approximate Value Iteration (Agostinelli et al., 2019) with a depth curriculum (Arfaee et al., 2011): $J(s) = 0$ if solved, else $\min_a [1 + J_{\text{target}}(s')]$, with targets from a network frozen every 100 steps. Identical architecture (600–4096–2048–1024–1, 12.95M parameters), batch 1024, Adam at 10^{-3} , on every rung. Three seeds throughout; a seed sets both the scramble stream and weight initialisation. Evaluation scrambles use a separate fixed seed held constant across configurations, so any two checkpoints are compared on identical states. Every solution is replayed on the puzzle before being counted.

Sampling. Profiles draw states across scramble lengths 1..16 and then *measure* the true distance. Drawing from deep random walks (the obvious choice, since it approximates the uniform distribution) concentrates almost every state into the shells nearest the mean, leaving the shallow half of the profile unmeasured. Scramble length is never used as the label: a state scrambled nine moves is usually nearer than nine.

4 Ordering accuracy decays with distance, for every informative heuristic

For each true-distance shell d we measure $\Pr[h(s) < h(s')]$ for s at distance d and s' at $d + 1$: can the heuristic still tell closer from further? At 1.0 it orders the two shells perfectly; at 0.5 it is a coin flip and search at that depth is blind, however small its regression error.

The comparison, and the baseline that broke it. We began by comparing the DAVI-trained network against one pattern database (the exact distance table of a smaller rung, admissible on the larger by abstraction) and a uniform random control. On rung $k = 6$ the learned heuristic decayed and the abstraction was nearly flat, which we read as a signature of goal-anchored bootstrapping. That reading does not survive a stronger baseline. The abstraction we used was rung $k = 2$, whose pooled GDRC on $k = 6$ is $+0.570$; a heuristic that weak is flat because it has little accuracy to lose, not because abstraction confers distance-invariance. Table 2 reports every abstraction available on that rung instead of one.

Controlling for strength. Decay is a function of how good a heuristic is. Across 39 distinct heuristics (22 learned checkpoints, 12 abstractions, 5 random controls) it correlates with pooled GDRC at $r = +0.588$ ($p = 8.3 \times 10^{-5}$; Figure 2). Regressing decay on GDRC and comparing residuals, learned heuristics sit $+0.006$ above the fit and abstractions -0.016 below it, a difference of $+0.023$ that is not detectable (Welch $t = +0.72$, $p = 0.48$) and whose 95% confidence interval, $[-0.039, +0.084]$, is small relative to the mean decay of 0.154. The strictest form of the test matches each learned heuristic to the abstraction nearest it in GDRC *on the same sampled states* (median gap 0.057): across 18 such pairs the abstraction decays more in 13, and the two mean decays agree to three decimals at $+0.154$.

heuristic	pooled GDRC	acc. at $d=1$	at $d=6$	decay
random	+0.003	0.495	0.505	-0.010
PDB $k=2$	+0.570	0.747	0.603	+0.144
PDB $k=3$	+0.685	0.834	0.628	+0.206
PDB $k=4$	+0.788	0.888	0.663	+0.225
learned	+0.842	1.000	0.801	+0.199
PDB $k=5$	+0.891	0.990	0.712	+0.278

Table 2: Rung $k = 6$, all heuristics on identical states and identical exhaustive ground truth, ordered by pooled GDRC. Decay grows with strength across the column, and the steepest decay belongs to a pattern database, not to the learned heuristic.

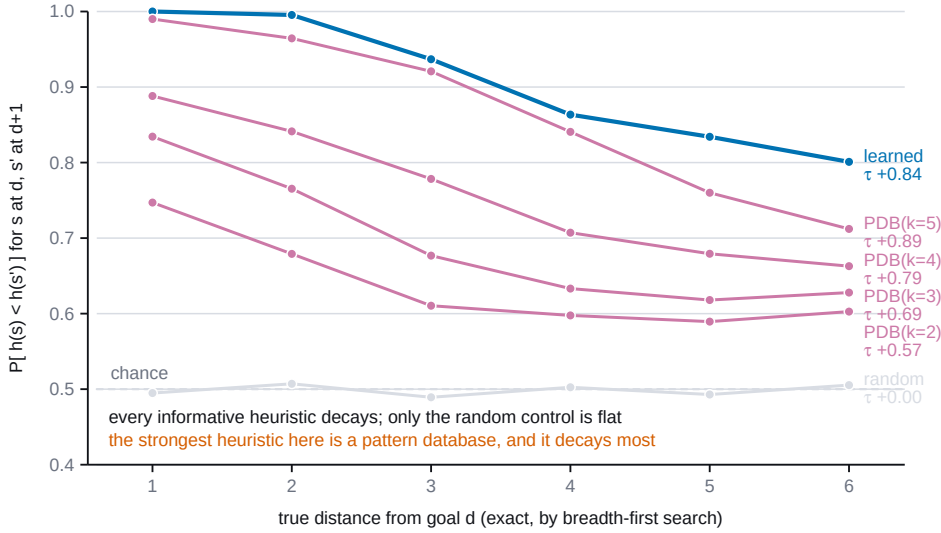


Figure 1: Ordering accuracy by true distance on rung $k = 6$. Every informative heuristic is near-perfect adjacent to the goal and decays outward; only the random control is flat. Showing PDB($k=2$) alone, the lowest curve above chance, is what made the decay look particular to the learned heuristic.

What does survive. The profile itself, and its dependence on depth. The decay grows with diameter at *constant* cardinality of 255,024 states: 0.108 ± 0.008 at diameter 6, 0.123 ± 0.006 at 8, and 0.259 ± 0.029 at 10 (mean over three seeds). On rung $k = 8$ (2.97×10^{10} states), where ground truth comes from the verified prover, learned accuracy runs 1.000 at $d = 1$ to 0.677 at $d = 6$. So there is a real distance-dependent structure to measure, in a heuristic class where it was expected and in one where it was not, and Section 7 is about the fact that a pooled coefficient reports none of it.

What we no longer claim. That the profile distinguishes learned heuristics from classical ones, or that its shape diagnoses bootstrapping. We report the earlier claim and its refutation rather than deleting it, because the error is instructive: the flat baseline was not evidence of a mechanism, it was an artifact of comparing against something barely better than chance.

5 The profile is a two-moment quantity

Section 4 leaves the decay described but unexplained. The explanation is one sentence: a heuristic tells two shells apart exactly as well as the gap between their average values is large compared with the scatter of values inside them. If the averages differ by one move and values scatter by a tenth of a move, the two shells barely overlap and ordering is near-perfect; if they scatter by a full move, the shells overlap heavily

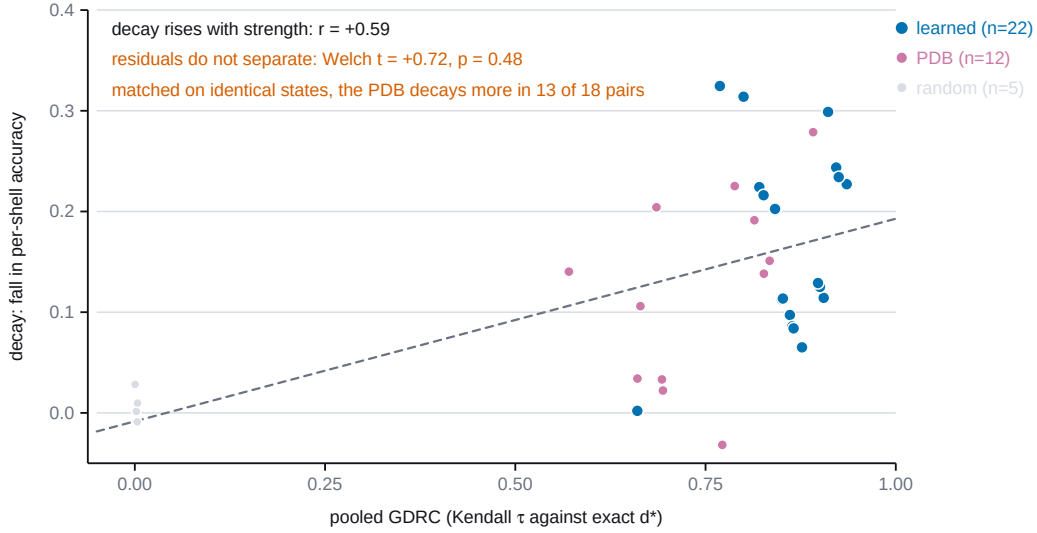


Figure 2: Decay against pooled GDRC for every distinct heuristic measured. The dashed line is the least-squares fit. Learned points do not sit above it. Repeats are collapsed: a profile is written per learned checkpoint and re-measures the same abstractions on the same states, so counting each appearance separately would inflate n from 39 to 98.

heuristic class	n	mean GDRC	mean decay
random	5	+0.003	$+0.003 \pm 0.014$
PDB	12	+0.741	$+0.125 \pm 0.095$
learned	22	+0.860	$+0.171 \pm 0.085$

decay vs. strength: $r = +0.594$ over 39 observations ($p = 6.7 \times 10^{-5}$)
residual about that fit: learned +0.0062, PDB -0.0162; Welch $t = +0.72$, $p = 0.48$
difference +0.0224, 95% CI $[-0.039, +0.084]$
matched on identical states: PDB decays more in 13 of 18 pairs

Table 3: The strength control in full.

and ordering approaches a coin flip. Nothing about the heuristic’s provenance enters. Formally: Write $\mu(d)$ for the mean heuristic value on shell d , $\sigma(d)$ for the within-shell spread, and

$$\Delta(d) = \mu(d+1) - \mu(d), \quad d'(d) = \frac{\Delta(d)}{\sqrt{\frac{1}{2}(\sigma(d)^2 + \sigma(d+1)^2)}}.$$

If the two shells’ value distributions were normal with equal variance, the probability that a draw from the nearer shell falls below a draw from the further one would be exactly $\Phi(d'/\sqrt{2})$. We find that this predicts the measured ordering accuracy across our data:

$$a(d) = \Phi\left(\frac{d'(d)}{\sqrt{2}}\right). \quad (1)$$

Nothing here is fitted. There is no coefficient, no intercept, and no per-domain constant, so Equation 1 is a prediction that could simply be wrong, most obviously for pattern databases, whose values are small integers with heavy ties and no reason whatever to look normal.

The estimate must not use the states it is scored on. Accuracy and the two moments are both computable from the same sampled shell, and if they are, the comparison is partly circular: a shell whose

sample	n	mean error	noise floor	max error
exhaustive ground truth, $k \leq 6$	491	0.0269	0.0019	0.314
<i>same, moments and accuracy from one sample</i>	491	0.0202	0.0019	0.231
<i>held out</i> : probed ground truth, $k=8$	6	0.0078	0.0024	0.017
<i>held out</i> : other training objectives	232	0.0166	0.0020	0.161
Two free parameters, $\Phi(d'/s + b)$, give mean error 0.0248 at $s = 1.393$, against 0.0202 at the parameter-free $s = \sqrt{2} = 1.414$				

Table 4: Equation 1 against measurement. Each observation is one heuristic at one true-distance shell.

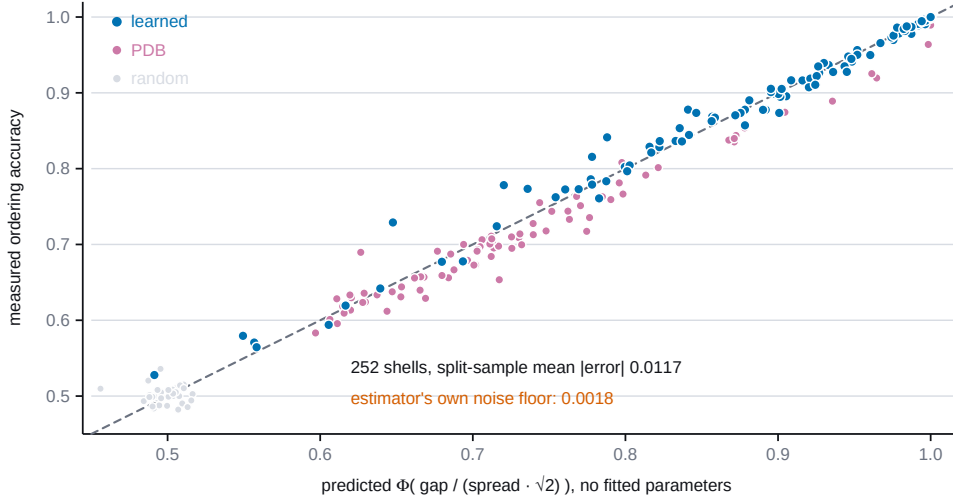


Figure 3: Predicted against measured ordering accuracy; the dashed line is $y = x$. Pattern databases sit slightly below it, which is the cost of their discreteness, the one place the normal approximation is visibly imperfect.

sampled mean happens to sit high yields both a higher predicted accuracy and a higher measured one. The size of that effect is not small. Run this estimator on a heuristic with *no signal at all* and it reports a correlation of +0.95 between prediction and measurement where the truth is zero. We therefore split every shell, estimate Δ and σ on one half and measure accuracy on the other, and report the split figures throughout. The correlations survive for informative heuristics and collapse for the random control, which is the expected signature, and is why we do not quote a correlation coefficient as evidence anywhere in this paper.

Table 4 and Figure 3 report the comparison. Split-sample, the mean absolute error is 0.012 over 252 shell observations. That is roughly six times the noise floor of the estimator itself (0.002, from the binomial error of accuracies computed on 20,000 sampled pairs), so the residual is real structure rather than measurement noise: the law is a good approximation, not an exact one. It holds on rung $k = 8$, where ground truth comes from the meet-in-the-middle prover rather than exhaustive search (0.008), and on 152 shell observations from heuristics trained under three objectives introduced after this measurement was designed (0.015). The error is largest for pattern databases, whose small-integer values are the furthest from normal.

Is the form right, or merely monotone? Any increasing function of d' would track the measurement, so agreement alone is not the test. Granting the model two free parameters, $\Phi(d'/s + b)$, improves the mean absolute error only from 0.0087 to 0.0082, at $s = 1.451$ against the $\sqrt{2} = 1.414$ the identity specifies. A fitted logistic does worse (0.010). We note that $\sqrt{2}$ is not an empirical discovery: it appears because

noise family	tails	n	measured	forward
uniform	lighter than normal	6	0.0042	0.0065
gaussian	normal (the assumption)	6	0.0046	0.0079
laplace	heavier, finite variance	6	0.0123	0.0458
cauchy	no finite variance	6	0.1654	<i>undefined</i>
Mean absolute error of Equation 1, over $\lambda \in \{0.25, 0.5, 1.0, 2.0, 4.0, 8.0\}$ in units of the unperturbed within-shell spread. <i>Forward</i> uses only the $\lambda=0$ moments.				

Table 5: Injecting calibrated noise into a trained heuristic on rung $k = 6$. *Measured* recomputes both moments from the perturbed heuristic; *forward* predicts accuracy from the unperturbed moments plus the analytic variance inflation $\lambda^2 \text{Var}(\varepsilon)$, touching the perturbed heuristic only to score it.

we write the denominator as a root-mean-square of the two shells’ spreads, and $\Phi(\Delta/(\sigma_{\text{rms}}\sqrt{2}))$ is identically $\Phi(\Delta/\sqrt{\sigma_1^2 + \sigma_2^2})$, which is exact for independent normals. What the fit tests is whether real value distributions behave like that, and they do, to within about 3% in the effective scale.

Breaking it on purpose. Everything above is observational: we measured heuristics that happened to exist. That leaves two objections. The comparison could be confounded, since our heuristics differ in rung, diameter, network and training. And it could be vacuous, since any increasing function of d' would track the measurement.

Both are answerable by manufacturing heuristics with known perturbations. Take a trained network and add deterministic per-state noise, $h_\lambda(s) = h(s) + \lambda\varepsilon(s)$, with ε derived by hashing the state’s index so that h_λ remains a function of the state. Every point in the sweep then shares rung, diameter, network and training exactly, which disposes of the first objection. The second is settled by the choice of noise family, because Equation 1 has no right to work when the perturbation has no variance to speak of.

Table 5 reports the outcome, and it is the sharpest evidence in this paper. The error grows monotonically with the tail weight of the injected noise: uniform and Gaussian noise leave the prediction intact (0.004–0.005), Laplace noise degrades it about threefold, and Cauchy noise, which has no finite variance, breaks it by a factor of 36, to 0.165. A law that were merely monotone in d' would have no reason to care which distribution the noise came from. This one does, in the direction and roughly the proportion the normal approximation implies.

The *forward* column is the causal test. Adding independent noise of variance v inflates each shell’s variance by exactly $\lambda^2 v$ and leaves the gap alone, so the unperturbed moments alone predict the perturbed accuracy with nothing fitted and nothing measured on the perturbed heuristic. That prediction holds to 0.008 on average and 0.013 at $\lambda = 8$, where the injected noise is eight times the heuristic’s own within-shell spread. For Cauchy the forward prediction cannot even be written down, which is the same fact stated twice.

Three consequences. First, d' is a ratio, so it is invariant to rescaling: multiplying a heuristic by a constant moves Δ and σ together and leaves $a(d)$ untouched. No amount of recalibrating output magnitudes can improve ranking, which also disposes of the span-based diagnostic we withdrew in Section 7. Second, the decay of Section 4 is not a consequence of a heuristic being good or bad: holding σ constant across shells makes $a(d)$ constant and the decay exactly zero, whatever the overall accuracy. Ordering quality falls with distance if and only if within-shell spread grows with distance, and it does for every heuristic we measured. But it does so in strikingly different ways, and an earlier version of this work got that wrong: pooling all heuristics gave $\sigma(d) \approx \sigma_0(1 + 0.21d)$, which we reported as a property of the state space. Separated by class, the two do not resemble each other at all. Abstractions grow gently from a wide base ($\beta = 0.15$, $\sigma_0 = 0.40$, $n = 11$); learned networks grow steeply from a narrow one ($\beta = 2.04$, $\sigma_0 = 0.04$, $n = 21$), with no overlap in range. The pooled figure was the abstraction value, surfacing because duplicated rows gave abstractions three times their true weight.

objective	solve rate	decay	beam width	curriculum
squared error (baseline)	100.0%	0.219	24.0	40
pairwise rank, uniform pairs	59.8%	0.354	302.2	33
pairwise rank, adjacent pairs only	44.3%	0.145	131.3	37
+ variance penalty, $\lambda = 1$	100.0%	0.223	21.8	40
$\lambda = 3$	100.0%	0.247	18.3	25
$\lambda = 10$	99.5%	0.417	49.2	7
$\lambda = 30$	92.0%	0.471	79.2	10

Table 6: Four training objectives on rung $k = 6$, identical in architecture, budget, seeds and curriculum criterion. Beam width is the smallest reaching a 90% solve rate; “curriculum” is the scramble depth the run reached, out of 40. Three seeds each; runs that never left the first curriculum level are excluded.

The corrected statement is more interesting than the one it replaces. The two classes arrive at comparable decay from opposite directions: a pattern database starts imprecise and degrades slowly, a learned network starts sharp and degrades fast, which is why matching them on overall strength makes their decay indistinguishable while their mechanisms are not.

Third, and most usefully, adjacent shells are one move apart, so Δ is pinned near 1 for any heuristic calibrated to distance; we measure 0.917 for learned heuristics and 0.600 for abstractions, whose values compress because abstraction shortens distances. The only remaining free term is σ . A squared-error objective charges bias and variance alike; ordering is indifferent to bias and cares only about variance relative to a gap it cannot enlarge. Section 6 tests whether an objective that targets that term directly yields a better heuristic. It does not, and the way it fails is itself informative.

What this is and is not. Equation 1 is not new and we claim no part of it. It is the equal-variance binormal identity relating d' to the area under an ROC curve (Hanley & McNeil, 1982), which descends from Thurstone (1927); the observation that per-shell ordering accuracy is a pairwise-class decomposition of AUC is Hand & Till (2001). Nearest in our own setting, Czechowski et al. (2021) model a learned value function as distance-plus-Gaussian-noise and report ordering-inversion rates against step separation, which is the same mechanism argued from the same premise. What we report is that it holds, parameter-free and to about one percentage point, for heuristic ordering quality measured against exact ground truth, across learned networks, integer-valued pattern databases and random controls, on every rung and diameter we can measure. Its value is that it reduces a profile to two interpretable numbers and identifies which of them an intervention can move.

6 Trying to flatten the profile makes the heuristic worse

Section 5 identifies within-shell spread as the term an objective can act on. We tried to act on it, and report the outcome in full because it is negative in an informative way.

We compare squared error against a RankNet-style pairwise logistic, the same loss restricted to pairs one shell apart, and squared error plus an explicit within-shell variance penalty at four weights. All arms share a scale-invariant curriculum criterion, because the default criterion advances on squared error and a scale-free ranking loss can never satisfy it, and using it would have pinned those arms at the first level and manufactured the result.

Nothing beat the baseline. At $\lambda = 1$ the variance penalty is indistinguishable from squared error on every measure (decay 0.223 against 0.219, $p = 0.77$; beam width 21.8 against 24.0, $p = 0.70$). Both ranking arms are substantially worse, and both are unstable: their outputs are scale-free, and across seeds the range of network outputs spanned 16 to 2817.

Pushing harder makes it worse, monotonically, and we can see why. Raising the penalty weight moves the decay in the wrong direction with perfect rank consistency ($\rho = +1.00$ across $\lambda \in \{0, 1, 3, 10, 30\}$), and the mechanism is visible in the last column of Table 6: the curriculum level reached falls with λ ($\rho = -0.87$). Penalising within-shell spread makes the network’s outputs more nearly constant, a curriculum gated on adjacent-shell separation reads that as failure to learn the current level, and the run never reaches the depths whose ordering the penalty was meant to protect. At $\lambda = 10$ training stalls at scramble depth 7 of 40.

The flattest profile belongs to the worst heuristic. The adjacent-pairs ranking loss produces exactly the effect Section 5 predicts. Restricted to the comparisons a search actually makes, it yields the flattest profile we measured, decay 0.145 against the baseline’s 0.219, it solves 44% of scrambles against the baseline’s 100%. Every arm here that succeeded in flattening the profile also degraded the heuristic, and the two effects are not separable in our data.

What we take from this. The decay is not a defect waiting to be removed. Within-shell spread is not noise contaminating a signal: it is how a heuristic distributes its predictions over depth at all, and suppressing it suppresses the thing being measured. A reader who takes Equation 1 as a design target should read this section first. We do not claim the space of interventions is exhausted, since targeting the numerator, or the variance of the bootstrap target rather than of the network, remain untried, only that the most direct reading of the law does not work, and fails for a reason the law itself does not anticipate.

7 What a pooled coefficient hides

The dilution can be stated exactly rather than argued. Pairs tied in d^* are precisely the pairs drawn from the same shell, so with n_s sampled states at distance s and $a(s, t)$ the probability that h ranks a state at s below one at t ,

$$C - D = \sum_{s < t} n_s n_t (2a(s, t) - 1), \quad n_0 - n_1 = \sum_{s < t} n_s n_t,$$

so τ is an $n_s n_t$ -weighted average of ordering accuracy over *every* pair of shells. Adjacent shells are a vanishing share of that sum. A search never makes the other comparisons: beam search, greedy best-first and A* all compare a node against its siblings, one move apart and therefore in adjacent shells. **The comparison a search depends on is the hardest one, and it is the one τ dilutes most.** Combining this identity with Equation 1, which supplies $a(s, t)$, predicts the pooled coefficient from the profile alone to within 0.016 for learned heuristics, so the profile, not the scalar, is the more fundamental object.

One run of the forty in Section 8 degenerated: the deadlocked one, which trained only within two moves of the goal. It is used here as a case study in what a pooled coefficient conceals, not as evidence about depth. Its pooled τ is +0.660, comparable to the abstractions available on those same states (PDB $k=2$ at +0.693, PDB $k=3$ at +0.772) and far below its sibling seeds at +0.936. Those PDB figures are inflated by ties: corrected as in the next paragraph, PDB $k=2$ scores +0.618, so a pooled τ ranks this crippled network *above* a working pattern database. Its per-shell profile shows why: adjacent-shell accuracy falls to 0.498 by $d = 9$ while the pooled coefficient stays near +0.66, because that coefficient is dominated by the far-apart pairs the identity above weights most heavily. It orders distant states well and adjacent ones barely above chance; τ reports the average, while search experiences only the adjacent column.

τ -b also rewards a heuristic for being coarse. Kendall’s τ -b divides by $\sqrt{(n_0 - n_1)(n_0 - n_2)}$, where n_1 counts pairs tied in the heuristic. A heuristic with no ties is charged for its arbitrary *within*-shell ordering, and one that ties inside each shell is not. On our $k = 6$ sample a tie-free heuristic that orders every shell perfectly is capped at 0.919, while the exact distance function, which makes the identical shell ordering but is integer-valued, scores 1.000. Both are computed analytically and checked against `scipy`, which agrees to four decimals.

The gap is not a curiosity. Pattern databases are small integers; learned heuristics are continuous. Breaking each abstraction’s ties at random, which leaves its between-shell ordering completely unchanged, costs it

generators	#moves	diameter	seeds	solve rate	excl. deadlocked
all	63	6	10	100.0 ± 0.0	–
oi-q	24	8	10	100.0 ± 0.0	–
oi-q4	16	10	10	100.0 ± 0.0	–
oi-q3	12	12	10	97.3 ± 8.0	100.0 ± 0.0
1 of 40 runs never advanced past the first curriculum level; see Section 8					

Table 7: Rung $k = 4$ at four generating sets. Every row has exactly 255,024 reachable states, verified by exhaustive search; only the diameter differs. Ten seeds each, identical 12,000-step budget.

0.070 to 0.101 of pooled τ across our configurations (mean 0.084). That is the same magnitude as the differences the learned-versus-classical literature reports from this statistic. Comparing the two classes by GDRC is therefore not a like-for-like comparison. That τ -b is sensitive to ties is Kendall (1945)’s own observation and not a new one; what appears not to have been noted is that it is large enough to reverse the learned-versus-classical comparisons this statistic is used to make in search. The remedy is one line: break ties at random before correlating, report τ both ways, or use an asymmetric measure such as Somers (1962)’s D , which conditions on the variable that has the ties.

A correction. We first attributed this to τ being scale-invariant, since that run’s outputs span only 0.004. That explanation is wrong: a heuristic with a small span and clean ordering works perfectly well, and Wilt & Ruml (2016) rely on scale-invariance deliberately. What degrades the run is that its within-shell spread ($\approx 6 \times 10^{-4}$) exceeds its between-shell gap ($\approx 2 \times 10^{-4}$), so adjacent shells overlap. Span is not a diagnostic and we do not propose it as one.

Solve rate is not a substitute. That same degenerate run still solved 73.5% of its evaluation scrambles. With 12 generators at beam width 100 over depth 60, the search touches an appreciable fraction of a 255,024-state space: this is beam search brute-forcing the problem, not the heuristic contributing.

8 Neither depth nor size, in this range

Table 7 holds cardinality exactly constant and varies only the generating set. Doubling the diameter from 6 to 12 changes the solve rate by nothing measurable: every seed at every diameter solves every evaluation scramble, with the single exception discussed below. Cardinality is ruled out by construction, and depth is not doing anything either over this range.

A correction, and how we found it. An earlier version of this section reported that performance “breaks at diameter 12”, on the strength of three seeds of which one collapsed. Checking the final curriculum level of every checkpoint showed that the collapsed run had never left the first level: its network settled to a constant at scramble depth 2, its squared error therefore never fell below the threshold that advances the curriculum, and it spent all 12,000 steps training on states within two moves of the goal. It did not fail at depth; it never trained at depth. Re-running the experiment at ten seeds per diameter puts that failure rate at 1 in 40, and the failure is not associated with diameter (1 of 10 at diameter 12 against 0 of 30 elsewhere, Fisher exact $p = 0.25$). With that run removed, every cell in the table is 100.0 ± 0.0 .

We report this because the failure mode generalises past our own experiment. A curriculum gated on a loss threshold is a feedback loop: a run that fails to fit the current level is denied the deeper data that might have dislodged it. The resulting collapse looks exactly like a hard instance (low solve rate, tiny output range, near-chance ordering far from the goal) and is nothing of the kind. The diagnostic is one integer, the curriculum level reached, and it costs nothing to record.

What we still claim here. That state-space cardinality is not the operative variable, which the design establishes by holding it exactly constant. Not that depth is: this range shows no depth effect at all, and a

k	states	solve rate at training budget		
		3k	12k	40k
2	5.52×10^2	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
4	2.55×10^5	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
6	9.69×10^7	93.5 ± 5.8	100.0 ± 0.0	99.8 ± 0.2
8	2.97×10^{10}	16.3 ± 14.4	81.3 ± 7.0	93.3 ± 4.9
10	7.12×10^{12}	1.2 ± 0.8	20.7 ± 11.1	39.0 ± 23.5
12	1.30×10^{15}	0.3 ± 0.5	10.8 ± 7.0	22.3 ± 10.1
16	1.54×10^{19}	0.0 ± 0.0	5.3 ± 2.6	10.7 ± 3.5
24	3.10×10^{23}	0.5 ± 0.0	1.2 ± 0.8	0.7 ± 0.9

Table 8: Solve rate (%) across 21 orders of magnitude of state space, three seeds per cell, mean $\pm$ sd. Branching factor, encoding, architecture and budget are identical throughout. Note that variance peaks in the transition region.

design that varies depth and branching factor together could not have attributed one anyway. Seed variance across the *size* ladder (Table 8) is a separate matter and is unaffected, because every run there reached the curriculum ceiling, so its ± 23.5 percentage points at $k = 10$ is genuine variation rather than this deadlock.

9 A second domain, and where the law stops

Every measurement so far is a Rubik’s-cube sub-problem, so every claim above is one state space seen through several sub-problems of itself. This section adds the sliding-tile puzzle: the 3×3 board, which is the 8-puzzle, and the 2×4 . Exact distances come from breadth-first search over a Lehmer ranking of the board. The implementation is checked against numbers it cannot influence: it reproduces $n!/2$ reachable states on every board, the published diameters of 6, 21, 36 and 31 for 2×2 , 2×3 , 2×4 and 3×3 , and the known count of exactly two states at distance 31 on the 8-puzzle.

Abstractions are built as they are on the cube: track the blank and tiles $1..j$, forget the rest. The measurement code is not duplicated. Both domains go through one implementation of shell binning, pair sampling and tie counting, because two scripts that merely resemble each other would let any difference between them surface as a domain effect.

One trap is worth recording, because it would have flattered the classical baseline rather than harming it. On the 8-puzzle, $j = 6$ leaves two tiles untracked, and the parity invariant then fixes which way round those two go, so the abstraction still separates every state and its table *is* the exact distance. As a baseline it scores 1.000 at every shell and a GDRC of exactly one. That is an oracle, not a strong heuristic. We exclude any rung whose reachable count has not fallen below the full task’s.

The profile results replicate. On the 8-puzzle the learned heuristic reaches GDRC +0.965, with adjacent-shell accuracy falling from 1.000 to 0.711 across the measured shells. The nearest abstraction in strength, $j = 5$ at GDRC +0.995, falls from 1.000 to 0.690: slightly stronger overall and decaying slightly *more*. The random control stays flat at chance. Decay with distance is not a fingerprint of bootstrapping in this state space either, which is the claim of Section 4 surviving a change of domain.

The law is worse here. Split-sample mean absolute error is 0.0117 over 252 cube observations and 0.0429 over 239 sliding-tile observations, against an estimator noise floor near 0.002 in both. Read as a domain split that says Equation 1 does not generalise, and we first wrote it down that way.

It is not a domain fact. What the error tracks is the skewness of the within-shell value distribution, Spearman +0.74, and the relationship is monotone across all three tasks (Table 9). Matched on skew, the domain gap disappears below $|\text{skew}| = 1$, where the tile fits slightly *better* than the cube in every bin. Reweighting the cube to the tile’s skew composition moves it only from 0.0221 to 0.0237, nowhere near the

task	n	median $ \text{skew} $	mean $ \text{error} $	bias
cube, $k=6$	24	0.599	0.0221	-0.0220
sliding tile, 3×3	64	0.858	0.0508	+0.0465
sliding tile, 2×4	46	1.557	0.0718	+0.0666
<i>Matched on skew, the domain gap closes below $\text{skew} = 1$:</i>				
$ \text{skew} $ range	cube n	cube $ \text{err} $	tile n	tile $ \text{err} $
0 to 0.4	5	0.0089	29	0.0065
0.4 to 0.7	9	0.0199	12	0.0138
0.7 to 1	2	0.0272	15	0.0243
1 to ∞	8	0.0315	54	0.1080
Reweighting the cube to the tile’s skew mix gives 0.0237, against the tile’s 0.0595.				

Table 9: The law’s error tracks within-shell skewness rather than the state space. Top: the three tasks in increasing order of median skew. Bottom: matched on skew, the cube and the sliding tile agree below $|\text{skew}| = 1$ and diverge above it. Pattern databases only, so this reproduces without a trained model.

heuristic class	cube		sliding tile	
	n	mean $ \text{error} $	n	mean $ \text{error} $
learned	138	0.0068	74	0.0126
PDB	80	0.0200	129	0.0667
random	34	0.0119	36	0.0203

Table 10: Where the domain gap lives. Every class is worse on the sliding tile, but pattern databases are worse by nearly twice the factor learned heuristics are, which is the split the skew account predicts.

tile’s 0.0595, which says the cube and the tile are not two populations with different laws but one population indexed by skew.

The class breakdown says the same thing from the other side (Table 10). Moving to the sliding tile costs learned heuristics a factor of 1.9 in mean absolute error and pattern databases a factor of 3.3. A learned value function is smooth and close to symmetric within a shell; a coarse abstraction is not, and the sliding tile’s abstractions are the more skewed of the two. That is why the cube alone could not have located this boundary: its abstractions never reach the skew where the law begins to fail.

Four explanations we tested and rejected. Three are backwards from the prediction, and we record them because a ruled-out explanation is the cheapest thing to hand a reader who would otherwise reach for it first. *Saturation*, that a weak abstraction tops out at its own small diameter: predicted the weakest rungs would misfit worst, measured Spearman +1.000 the other way, the strongest rung being the worst fit at 0.122 and the weakest the best at 0.035. *Near-determinism*, that a within-shell spike is the least normal shape available: error *falls* with d' , Spearman -0.53 , reaching 0.0003 above $d' = 5$. *A strength confound*, which is this paper’s own warning about unmatched baselines turned on itself and so had to be checked: reweighting the cube to the tile’s d' composition gives 0.0160 against the tile’s 0.0595, and within matched d' bins the tile remains several times worse. *Ties*, which the equal-variance binormal identity assumes away and which Section 7 shows biases τ -b: backwards again, since the cube has a median tie rate of 0.325 and the tile 0.000.

Equal variance, the assumption we named explicitly when deriving Equation 1, is not the culprit either. The tile’s variance ratio is 1.258 against the cube’s 1.428, better behaved, and reweighting on it closes almost none of the gap. The assumption that fails is normality, and specifically its third moment, which we had not singled out.

What remains unexplained. Above $|\text{skew}| = 1$ the tile is still 3.4 times worse than the cube at matched skew. Skewness accounts for the gap over most of the range and leaves a residual we cannot attribute. We report the boundary we can demonstrate, not a mechanism we cannot.

10 Limitations

Ground truth stops at $k = 8$. Enumeration is infeasible past $k = 6$. Meet-in-the-middle buys exactly one further rung. A $k = 10$ run was attempted and abandoned after 31 hours: its mean distance is ≈ 9.9 while a radius-6 ball with four backward levels reaches only distance 10, so roughly half its states would return unresolved, the deep half the profile exists to measure. This is the same memory wall that stops enumeration, arriving one rung later, and it is why rungs above $k = 8$ are reported by solve rate alone.

Depth and branching are not separated. Restricting generators reduces branching factor as it increases diameter. The design rules out cardinality; it cannot apportion the remainder.

We do not show that any of this makes a search cheaper. This is the most important limitation and we state it first. An earlier version of this work reported that the stratified measure predicts required beam width where pooled GDRC does not. It does not survive its own censoring: 11 of 19 configurations reach the target at beam width 1, and on the 8 where the outcome varies the ordering reverses in favour of pooled GDRC. We removed the section rather than report a correlation that inverts on the subset carrying the information. Establishing predictive value needs an outcome with no floor (nodes expanded at a fixed generous width, rather than the width required to hit a target) and a censored-data estimator. Until then the contribution here is diagnostic: it says what a heuristic’s ordering quality is made of, not what it will cost you.

The law is an approximation, and we can now say where it degrades. On the cube, split-sample mean absolute error is 0.012 against an estimator noise floor of 0.002, so the residual is six times the noise and Equation 1 is demonstrably not exact. Section 9 makes that statement sharper: the error scales with the skewness of the within-shell distribution, monotonically across three tasks in two state spaces, and reaches 0.072 where the median $|\text{skew}|$ reaches 1.56. The law should be used where the within-shell distribution is close to symmetric, which covers every learned heuristic we measured in both domains, and treated with suspicion on coarse abstractions, which is where the skew lives.

One family of abstractions. Every classical baseline here is the exact distance table of a smaller rung of the same ladder, so they form a nested sequence and differ from one another only in how many pieces they track. That is what makes them cleanly comparable in strength, and it is also the limit of the comparison: we show that *these* abstractions decay like learned heuristics of the same strength, not that all classical heuristics do. A heuristic built on a structurally different relaxation could behave differently.

Two domains, not many. Section 9 adds the sliding-tile puzzle, which is enough to show that the profile results are not a cube artifact and enough to locate a boundary on Equation 1. It is not enough to characterise that boundary. Two domains give three tasks and one monotone trend in skewness; they cannot say whether the residual above $|\text{skew}| = 1$ is a third-moment effect we have mismeasured or something else. Classical planning, where learned heuristics are also trained from goal-anchored data and where the state spaces are not permutation groups at all, remains untested.

A null result is not a proof of equality. Section 4 reports no detectable difference between learned and classical heuristics once strength is controlled. With 22 learned and 12 abstraction observations the test would not detect a difference smaller than roughly 0.08 in decay, so we claim that the training method is not the dominant term, not that it contributes nothing. The 22 learned observations also include intermediate checkpoints of the same runs, which are not fully independent of each other; the matched-pair test is restricted to final checkpoints for that reason.

What we do not claim. That a ranking statistic predicting search cost is novel (Wilt & Ruml, 2016); that predicting search cost from an intrinsic heuristic property is novel (Korf et al., 2001); that required budget is independent of state-space size, which Korf et al. (2001) contradicts; that the profile distinguishes learned heuristics from classical ones, which Section 4 refutes; that the profile predicts search cost better than a pooled coefficient, which our own censoring analysis refutes; novelty for Equation 1, which is classical; any admissibility guarantee; or a new solver.

11 Conclusion

A heuristic’s ability to order states varies with distance from the goal, and that variation is not mysterious: it is $\Phi(\Delta/(\sigma\sqrt{2}))$ in the between-shell gap and the within-shell spread, to about a percentage point, without fitting anything. Two numbers per shell reproduce the profile; the profile reproduces the pooled coefficient; the pooled coefficient reproduces neither.

That account settles a question we had originally answered wrongly. The decay is not a fingerprint of bootstrapping. It occurs exactly when within-shell spread grows with distance, which is a fact about the state space rather than about training, and pattern databases on the same states decay just as much. We expected the opposite and say so plainly.

It also says where to push, and we pushed there and failed. Adjacent shells are one move apart, ranking is invariant to rescaling, and the within-shell spread is the only term left, so we penalised it. At low weight that changes nothing; at high weight it is monotonically harmful, because suppressing spread stalls the very curriculum that builds it. The flattest profile we produced belongs to one of the worst heuristics we produced. We report this because the law reads like a design target and is not one: spread is not noise contaminating the signal, it is how a heuristic covers depth at all.

A second state space moved the claim rather than confirming it. On the sliding-tile puzzle the profile results replicate, pattern databases included, but the law’s error is nearly four times larger. That difference is not a property of the domain: it tracks the skewness of the within-shell distribution, monotonically across three tasks, and matched on skew the two domains agree below $|\text{skew}| = 1$. So Equation 1 is skew-limited, not cube-specific, and the boundary was invisible from the cube alone because cube abstractions are not skewed enough to reach it. We tested four other explanations, three of which came out backwards from the prediction, and a residual above $|\text{skew}| = 1$ remains unaccounted for.

The practical recommendations are narrow and cheap. Report the profile rather than the scalar. Match heuristics on overall strength before comparing their profiles, because a weak baseline is flat for reasons that have nothing to do with the mechanism under study. Break ties at random before computing a rank correlation, or a coarse heuristic will beat a continuous one that orders states better. And record the curriculum level a run reached, because a training failure and a hard problem look identical without it.

Reproducibility statement

Every number in this paper is written by a script into a JSON file and read from there; none is transcribed by hand. Code, exact commands, pinned dependencies, runtimes and the seeds used, together with a list of superseded results (retained rather than deleted) and of measurements the apparatus could not obtain, are at

<https://github.com/acebot712/cubeworks>

One directory is deliberately absent. The detector’s real-frame evaluation set is photographs of a person and a home interior, so it is excluded from the repository permanently. No result in this paper reads it, the remaining tests pass without it, and `eval/README.md` gives the capture tool and the label schema for anyone wanting to rebuild an equivalent set.

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